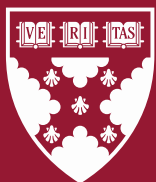


Working Paper 21-010

Analyzing Transformative Advances

Note on Productive Knowledge

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Business
School**

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* Amar Bhidé, now at Columbia University, first wrote this Note in 2021 when he served as a Visiting Professor at Harvard Business School. The subsequent 2025 revision is minor.

Note on Productive Knowledge

This Note examines the development of ideas ('knowledge') embodied in transformative products and services. It highlights the techniques and venturesome spirit of 'multiplayer' development – advances by and for the many – suggesting ways of using the techniques without harming the spirit.

By traditional and modern standards, actionable ('productive') knowledge embodied in useful products has low intellectual standing. The ancient Greeks venerated contemplation, music, other arts, abstract truths, and mathematical reasoning. Merchants and artisans occupied the bottom rung of Plato's idealized society; their knowledge and toil were but means for the good life of a small, enlightened class. Modern societies now venerate scientific knowledge. Engineers, physicians, lawyers, entrepreneurs, managers, and accountants earn high incomes. Still, many dismiss their knowledge as a mere application of deeper scientific principles.

Yet, developing actionable ideas and valuable products affirms our humanity. We are human because we create, not just because we think lofty thoughts. As mountaineers endure danger and pain to climb peaks "because they are there," innovators seek challenges in what is not there. Creating new things, far beyond bodily needs, engages distinctively human capacities. We imagine and reason, believe and question, compete and cooperate. We seek adventure and endure the anxieties that the adventures produce.

This Note focuses on—to put it colloquially—the massively multiplayer way we now develop transformative products. Although interconnected progress predates recorded history, the extent of our venturesome collaborations is new and transformative. Multiplayer innovation now gives unprecedented scope for individuals with diverse skills, capabilities, and backgrounds to exercise their imagination and initiative. It also requires and satisfies a widespread enthusiasm for venturesome (rather than merely conspicuous) consumption.

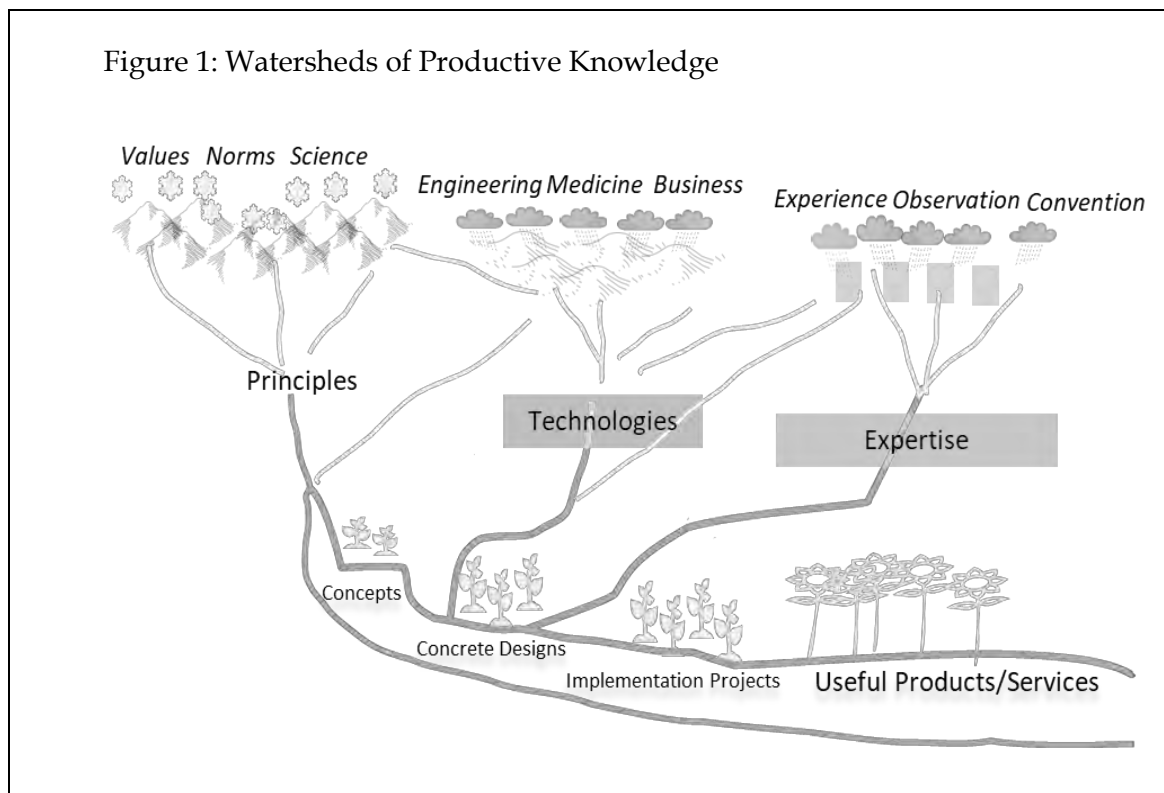
New managerial techniques help coordinate multiplayer advances. Although unplanned discoveries remain important, many new combinations ("ideas having sex," as Matt Ridley memorably puts it) now rely more on careful, selective breeding than on accidental or anonymous encounters. Silicon Valley has produced more than technological advances; companies like Intel have pioneered goal-setting systems to manage employees dispersed across diverse locations and functions. Yet, as we will see in this Note, misused techniques can induce mindless conformity and destroy the adventure of progress.

Watersheds of Productive Knowledge. Many technologies start with scientific discoveries. But just as much of the water that a river carries into the ocean does not originate in its headstreams, science does not provide

* Amar Bhidé, now at Columbia University, first wrote this Note in 2021 when he served as a Visting Professor at Harvard Business School. It draws from an earlier paper, "[Productive Knowledge: Sustaining Massively Multiplayer Innovation](#)."

all the knowledge embodied in the products we value. The headstreams and tributaries of the knowledge embodied in useful products (See Figure 1) include:

- **Principles**, often derived from science, social norms, and individual values. They consist of concise, broadly applicable propositions that can inspire useful high-level concepts.
- **Technologies** produced by engineering, medicine, business, and other practically oriented fields. The technologies—and related “techniques” and “practices”—have more detailed specifications but narrower scope than scientific principles. Contrast, for example Newton’s scientific laws of motion with the technology for designing distillation towers or techniques for performing arthroscopic surgery. The technologies, techniques, and practices help turn general principles into executable concepts and the concepts into concrete designs and procedures.
- **Expertise** acquired through personal experience or observation. Expertise is often tacit and contextual — best used in particular circumstances. It helps make designs workable and is essential to their implementation.



Innovators do not produce much of the knowledge they draw from the watersheds. Like the rain and snow from remote mountains, previously developed principles, technologies, and expertise provide free nutrients (“externalities”). And the proliferation of principles, technologies, and expertise has immensely expanded the possibilities for further transformative advances. Notably, the proliferation has included managerial techniques and protocols for coordinating broad-based, multiplayer innovation.

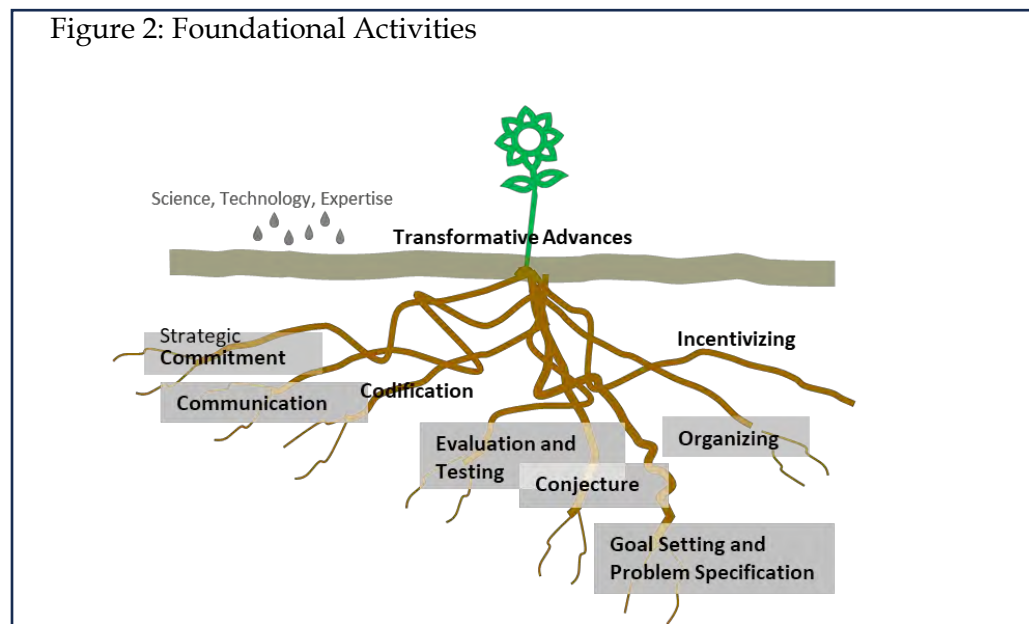
Yet, proliferation also poses problems. Just as too much mountain rain can flood downstream valleys, an overabundance of free knowledge can overwhelm users. Moreover, even fertile, unflooded fields don’t cultivate themselves. Gardeners and farmers must choose what to plant, how to fertilize, and when to

harvest. These choices require acquiring here-and-now (“local”) soil data, imagination, and leaps of faith. So does developing new products. Combining and extending existing ideas cannot be reduced to a mechanistic formula. Envisioning practical uses of scientific discoveries (such as magnetic resonance) requires imagination. Implementing designs and plans requires creative responses to unanticipated circumstances. Furthermore, bureaucratically used techniques and protocols to coordinate multiplayer development can suffocate the spirit of innovation.

Guide to a Guide. A National Geographic style guide map would help us choose and use suitable techniques from a bewildering assortment. But, constructing helpful guides requires deciding what information to provide and how. A river map, for instance, may have separate pages for the upstream headwaters, middle basins, and downstream deltas. Each page may show the main channels as smoothed, rather than photo-realistically jagged streams and display campsites but not the vegetation on the banks of the river. Guide maps also encourage visits to the places they cover. Even warnings about grizzly bears and rapids evoke alluring adventures.

This Note offers a template for a similar guide to innovative advances. It proposes eight foundational (“root”) activities for harnessing existing science, technology, and expertise. (Figure 2).

Figure 2: Foundational Activities



Specifically, the following three sections describe the:

- Primordial challenges, dating back to ancient times.
- Advantages and difficulties of modern multiplayer advances.
- Proposed eight-activity framework.

A concluding appendix contrasts the scientific and technological watersheds of productive knowledge.

I. Primordial Challenges

Willful Quest. Genetically encoded biological evolution provides an instructive contrast for the development of the knowledge embodied in manmade products. Like genetic information, the embodied knowledge is multifarious and multifunctional. For instance, making and selling a simple analgesic like ibuprofen requires knowledge spanning technical specifications (how many milligrams of active ingredient, binding agents, coatings, etc.), sourcing, manufacturing and quality control, logistics, packaging, advertising, and regulatory compliance. Just as genetic information evolves to encode more complex life forms – from single-celled organisms to humans – progress supports more progress. Time-keeping advanced from simple sundials to pendulum clocks, ship chronometers, and pocket watches. Moreover, as with genetic variations that produce a new species, the knowledge embodied in transformative products results from an extended accumulation of ideas, not a single bound.

But whereas time can make humans from prebiotic soup through purposeless mutations, transformative advances don't just randomly combine existing science, technology, and expertise. The protracted development of manmade things requires applying will, imagination, and reason (in addition to lucky accidents).

We choose short and long-term goals for the wants we'd like to satisfy, sometimes after accidentally stumbling into a possibility.

We form conjectures about how we might attain our goals and make willful choices about how to evaluate and test these conjectures.¹

Successful ideas diffuse through deliberate codification and communication. Even drilled or rote learning requires designing drills.

Commitments to strategic choices guide gradual development, leaving room for adjustments while maintaining a general overall direction. ² (See the Box 'Fixed-Wing Flight.')

Fixed-Wing Flight

The development of fixed-wing aircraft exemplifies persistence with strategic principles while adjusting tactics. Sir George Cayley first enunciated the underlying conjecture – that propelling a rigid surface through the resistance of air could produce an upward force ("lift") that would offset the downward pull of gravity – in 1809. All "airplane designers have this concept at the back of their minds" now, writes Walter Vincenti (a professor of aeronautical engineering at Stanford), but Cayley's concept was "revolutionary at the time" because it "freed designers from the previous impractical notion of flapping wings." Yet, it took nearly a century before the principle produced the first controlled flight of a powered, heavier-than-air aircraft on December 17, 1903, when the Wright Flyer took wing – for just 200 feet. In the interim, intrepid inventors experimented with gliders, steam engines, gasoline engines, propellers, automobile chains, and rudders. Otto Lilienthal, who had made the first well-documented, repeated gliding flights, broke his neck and died in 1896 after his glider stalled. Finally, the Wright Brothers built on these prior efforts, improved wing materials and designs, and pioneered the "three-axis" system to control flight.

Venturesome Leaps. Scientific principles, technologies, and expertise (from the “watersheds” shown in Figure 1) can help the willful innovator perform critical activities (of goal setting, forming and testing conjectures, and so on). But just as riverside farms don’t cultivate themselves, innovators cannot count on existing ideas to usefully combine themselves.

It’s obvious now that Cayley’s principle was sound and that the many failures that preceded the Wright Flyer reflected limitations of wing, airframe, propeller, and control designs. But efforts to develop fixed-wing airplanes, like alchemy, could have been a fantasy. Or, even if technically feasible, fixed-wing aircraft could have lost out to rigid airships, popularly known as “Zeppelins” (summarized in the Box ‘The Rise and Fall of Zeppelins’). Similarly, the synthesis of ibuprofen followed the screening of more than 600 compounds over more than ten years. This effort could have been futile, like attempts to cure the common cold.

The Rise and Fall of Zeppelins

Count Ferdinand von Zeppelin first formulated his idea for rigid airships in 1874. Over the next 20 years, he developed technical details, which he patented in 1895. After several failures and some fatal accidents, airships built by the Count’s eponymous Zeppelin Company were put into commercial service in 1910 by Deutsche Luftschiffahrts-AG (DELAG). DELAG, founded in 1909 by Count Zeppelin, thus became the world’s first revenue-generating airline. By the start of the First World War, DELAG had carried over 10,000 passengers in over 1500 flights.

Following the war, the Treaty of Versailles prohibited Germany from building large airships. After the restrictions were lifted in 1926, the Zeppelin Company started building the LZ 127 Graf Zeppelin. Work was completed in 1928, and the Graf (again operated by DELAG) began providing regular transatlantic commercial service in 1930. It was joined in 1936 by the larger LZ 129 Hindenburg. Unfortunately, in 1937, the Hindenburg caught fire in New Jersey after a transatlantic flight, killing 35 of the 97 people on board. The Graf Zeppelin was retired a month later. Thus ended the role of airships in providing commercially viable long-haul air transport that they, not fixed-winged airplanes, had pioneered.

But just as success isn’t certain, neither is failure. Invariably, protracted development poses “unmeasurable and unquantifiable risks.”³ Skeptics who bet against new technologies – producers of buggy whips, oil lamps, and sailing schooners – can be swept away.⁴

Therefore, those who persist—like those who don’t—make, to borrow from the 19th-century existentialist Søren Kierkegaard, ‘leaps of faith.’⁵ Moreover, those who make the first leaps also have to recruit others. Visionaries can rarely undertake pioneering innovations on their own and must have exceptional strong personal convictions to convince skeptics.

Thrift and bourgeois virtues of temperance and prudence celebrated by Max Weber and Deirdre McCloskey as the foundation of modern capitalism have their place, but only when joined to imprudent, against-the-odds audacity.

Consumers also cannot escape venturesome leaps. One simple reason is that different individuals have different tastes and preferences. A best-selling book may not delight all subsequent readers, and

patrons drawn to a three-star restaurant may leave disappointed. More subtly, unforeseen changes can make consumers' investments worthless. For instance, Sony's pioneering Betamax video format could not withstand the challenge of VHS, harming consumers who had accumulated libraries of Betamax videotapes. However, avoiding new technologies isn't safe either; buyers who stuck with sailing ships, like the shipyards that produced them, also lost out. Similarly, while experimental drugs can have dangerous long-term side effects, rejecting new diagnostic methods (to detect colon cancer, for instance) can be life-threatening.

Pragmatic Paradoxes and Combinations. Pragmatist philosophers such as Charles Sanders Peirce, William James, and John Dewey argue that the significance of ideas lies in their practical utility – “cash value,” as James puts it. Where Plato privileged truth that “lies in the abstract and exists more clearly in our minds than in the natural world,” the pragmatist asks what works rather than what is true.⁶

Developers of productive knowledge are clearly pragmatic in favoring the useful over the ultimately true. They also ‘paradoxically’ combine, as we will see next, ‘rationalist’ generalization with context-specific ‘empiricism,’ as well as progressivity with conservatism.

Rationalist Generalization + Context-Specific Empiricism. According to James, pragmatism conjoins the opposing dispositions of rationalists and empiricists. In James's classification, rationalists are “monists,” “devoted to abstract and eternal principles.” They “start from wholes and universals and make much of the unity of things.” Their truth lies (as in Plato) more clearly in the mind than in the natural world. Empiricists, in contrast, are “devoted to facts in all their crude variety” (see Box ‘Rationalists v Empiricists’). Like the fox in Isaiah Berlin's later essay, they seek to know many things, unlike the hedgehog, who knows one big thing.

Rationalists v Empiricists

The empiricists' world of “concrete personal experiences,” William James observed, “is multitudinous beyond imagination, tangled, muddy, painful, and perplexed.” In contrast, the rationalists' world is “simple, clean and noble. The contradictions of real life are absent from it. Its architecture is classic. Principles of reason trace its outlines, logical necessities cement its parts. Purity and dignity are what it most expresses.” But this latter world is just a “sanctuary in which the rationalist fancy may take refuge from the intolerably confused and gothic character which mere facts present. It is no EXPLANATION of our concrete universe, it is another thing altogether, a substitute for it, a remedy, a way of escape.”⁷

James's sympathies clearly tilt towards concrete empiricism, but crucially, he favors including broad abstractions when they do have practical utility. James's own pioneering work in psychology was not light on broad generalizations. Similarly, innovators must pay close attention to contextual facts in “all their crude variety” without discarding abstractions. Overhead bins of modern airplanes must be designed to accommodate roller carry-on bags. Similarly, producing a new aircraft requires knowledge of the quirks and capacities of specific manufacturing plants and suppliers. At the same time, developers of airplanes depend on the abstractions of fluid mechanics – and, as mentioned, commit to the principle of fixed-wing designs.

Progressivity + Conservatism. Pragmatism also balances tendencies that propel and restrain change. Nineteenth and early-20th-century pragmatists implicitly or explicitly embraced efforts to progress:

ultimate truths might never be discovered, but advances in knowledge that improved the human condition were always at hand.⁸

Yet in James's telling, pragmatic considerations require respecting existing ideas. James's pragmatist will seek out new ideas only to the degree that old ideas cannot deliver the goods and, even then, will favor adapting what exists rather than starting anew.

A similar combination characterizes useful innovation. A deep conviction that things can be made better, that dogged enterprise can overcome problems, nourishes the leaps of faith necessary to persist through setbacks. Yet, the existing stock of tangible and intangible capital and social and psychological conservatism favors retaining what is already known and used to whatever degree is possible.

Overwhelming Choices. Combining grand "monistic" leaps and myriad context-specific decisions creates tangles of choices. For instance, developing a self-driving vehicle raises questions about goals: What overarching purpose or purposes should the driverless vehicle try to serve? Reducing accidents, traffic congestion, driver stress, or labor costs? And, in what priority? Numerous specific goal-and-objective choices follow: about vehicle size, target cost, speed and range, reliability, and so on. Then, there are even more choices about means: navigational technologies, power sources (battery vs. gasoline), body materials, backup and monitoring mechanisms, production scale, financing, marketing, after-sales service, etc.

Simple trial-and-error provides limited help in making these choices. Consider [the extreme example Angus Deaton](#) offers of his four-year-old granddaughter using trial-and-error to master the popular Angry Birds game (played on mobile phones). The game has features that make trial-and-error effective: a simple goal (to kill as many pigs as possible), very few things players can manipulate, and immediate and unambiguous feedback.

These features are, however, absent in the development of most new products. As mentioned, development requires choices about goals and sub-goals—these are not simple 'givens.' Choices about means are complex, and the immediately apparent options are not the only ones potentially available. Developers of driverless cars, for instance, can choose an existing navigational technology or try to invent an alternative. Choices cannot be made one at a time: the target use for a driverless vehicle has implications for battery-technology choices. Furthermore, trials cannot provide immediate or unambiguous feedback. No tests can reliably anticipate the long-term, real-world performance of a new driverless vehicle.

If innovators could predict the consequences of all combinations of known and unknown options, of grand strategies and tactics, problems of real choice would not even arise. Like hydrogen combining with oxygen to produce water, we would do the foreordained. But purposeful choice is the essence of human existence. And choice entails anxieties that overwhelming combinations of options can make acute.

Certainly, energizing audacity mirrors dispiriting anxiety; known and unknown dangers provide the thrill and financial reward of leading transformations. But anxieties can also encourage "satisficing"—picking the first option that alleviates the problem at hand, and only when the situation becomes intolerable. Up to a point, such satisficing is the inevitable result, as Herbert Simon pointed out, of the "boundedness" of our rationality — our ignorance of all the options that might exist and of their

consequences. It is pragmatic to respect what's known to work ("if it ain't broke, don't fix it"), but that smothers our capacity for foresight, for making choices before we must, and for imagining options that do not naturally appear in front of us. Making do with what we have and know limits bold leaps and makes pragmatism more conservative than progressive.

II. Advantages and Difficulties of Multiplayer Advances

Transformative Widening. As mentioned, our capacity for collective advances has expanded vastly over the last 100 years. Looking back, we see that although many revolutionary products were invented between 1850 and 1900, they were usually developed by a few inventors. Alexander Graham Bell invented the telephone with one assistant. Automobile pioneers were one- or two-person shows: Karl Benz and Gottlieb Daimler in Germany, Armand Peugeot in France, and the Duryea brothers of Springfield, Massachusetts. But small outfits could not develop reliable products for mass consumption: early automobiles, expensive contraptions that broke down frequently, were purchased by wealthy enthusiasts "riding around the countryside terrifying horses."⁹

Innovation then became much more democratized starting in about the 1920s. The division and specialization of labor that dramatically increased manufacturing efficiencies in the early 20th century has now transformed the development of new technologies and products. The Internet, for instance, wasn't envisioned by a solitary Alexander Graham Bell. Innumerable entrepreneurs, financiers, executives of large companies, members of standard-setting institutions, researchers at universities and commercial and state-sponsored laboratories, programmers who wrote and tested untold millions of lines of code, and even investment bankers and politicians turned the Internet into a revolutionary medium of communication and commerce. Steve Jobs, often portrayed as a brilliant solitary inventor, relied on the contributions of tens of thousands of individuals working at Apple and its network of suppliers.

The broadening of venturesome consumption has provided crucial support for multiplayer development. Thomas Edison, the Wizard of Menlo Park, "devoted his talents to providing novelties for the urban upper class."¹⁰ Now, millions of the not-so-well-to-do line up to buy Apple's latest offerings. Larger demand also pays for the greater specialization of development. In innovation, as in Adam Smith's 18th-century pin factories, "the division of labor is limited by the extent of the market."

The venturesomeness of contemporary consumers also includes resourceful effort. Complex, feature-rich products – iPads and iPods included – usually don't "just work" out of the box. Producers cannot afford to provide individualized training and instead rely on resourceful consumers to learn about the hidden quirks of their novel offerings. Similarly, consumers modify products standardized for low-cost mass production to suit their individual needs. And some leading-edge consumers provide valuable suggestions and feedback to developers.¹¹

Role of Science and Technology. As scientific and technological research has increased, 'watershed' ideas available for innovative uses have multiplied. Science and technology have also subtly helped specialize and broaden knowledge work. For example, improved understanding of disease mechanisms has provided "targets" for new treatments. Additionally, targeting has helped pharmaceutical companies undertake high-volume, assembly-line screening of potentially therapeutic molecules. Science has thus helped expand the scope of drug discovery as well as industrialize the process. Similarly, miniaturized semiconductors have helped squeeze high-powered computers into mobile phones.

Concurrently, product design teams use print-on-demand and computer simulation technologies to test and refine prototypes systematically.

New radio, television, and internet technologies have helped create large markets that allow more specialization of innovative effort. High-tech mass marketing has supported large development teams at companies like Apple and Samsung who have taken talking to watches from fiction to reality. New materials and technologically supported specialization have also transformed traditional industries such as shoemaking (See Box, ‘Running on Air’).

Running on Air

Shoemaking was one of the first industries in the United States to specialize and automate production. By the early 20th century, affordable shoes, produced in large factories, made owning multiple pairs commonplace. Goodyear introduced “Keds” with vulcanized, treaded soles in 1892 but did not market them as athletic shoes until 1917. Adolf Dassler began making running shoes in 1920 for competitive runners: Jesse Owens won his Olympic gold medals wearing Dassler shoes.¹² But, these innovations did not launch the multiplayer development of running shoes for mass markets.

In 1960, New Balance Inc. introduced what is thought to be the first mass-produced running shoe, the *Trackster*. The *Trackster* was also the first shoe to be offered in varying widths, increasing its appeal. Once Nike pioneered waffle-soled shoes in 1972 and the Brooks Manufacturing Company introduced shoes to control foot rotation,¹³ one product innovation quickly followed another: shoes with proprietary cushioning systems (starting with Nike’s Air shoes) and pumps (pioneered by Reebok) as well as minimalist, ultralight shoes weighing less than 3 ounces. Advertising campaigns and endorsement contracts secured global brand recognition and billions of dollars in revenue for shoe companies. Outsourcing to factories in low-wage locations kept production costs in check. All this required specialists who once had no place even in “industrialized” shoemaking: bio and software engineers, material technologists and scientists, and designers; lawyers to negotiate endorsement contracts with sports agents; advertising agencies to produce commercials and purchase TV spots; and, supply chain professionals to manage outsourcing.

Managerial Advances. New organizations and their administrative routines have helped specialize and broaden innovation. Over the first half of the 20th century, 19th-century inventions such as automobiles moved from workshops (of pioneers like the Duryea brothers) to functionally organized, founder-controlled concerns (such as Ford Motor) to professionally managed multi-divisional corporations (such as General Motors). The new organizations didn’t simply house low-cost, high-volume manufacturing; they combined the contributions of many specialists—in industrial engineering, design, financial analysis, marketing, and logistics, for instance—to give consumers ever-new yet affordable products. In medicine, diverse teams (including researchers, clinicians, engineers, technicians, and publicists) employed by new multi-specialty private practices (such as the Mayo Clinic and the Cleveland Clinic) played pivotal roles in developing and disseminating treatments such as cardiac surgery. New kinds of professional firms employing diverse specialists (such as Arthur D. Little and McKinsey & Company) advanced new technical and managerial ideas. Mass discounters (such as Wal-Mart), multinational advertising agencies (such as McCann Erickson), and now e-tailers (such as Amazon) whet and fed appetites for venturesome consumption.

Organizational innovators developed more than just new kinds of organization charts. They also developed formal and informal routines for efficiently managing existing operations as well as undertaking new initiatives. Sophisticated project management techniques emerged as essential tools for undertaking complex megaprojects from developing Nvidia's semi-conductors to Amazon's same-day-delivery supply chains.

Mavericks continue to flourish, however. Spontaneous self-ordering – or managerial routines – did not create the Mayo Clinic and Wal-Mart. Visionaries like Jeff Bezos have continued to shape the development of behemoths like Amazon. Audacious medical researchers continue to lead transformative advances in immuno-therapies. But maverick pioneers do not act alone – they play in a broader, multiplayer game. The immunotherapy pioneer Michel Sadelain found a home at Memorial Sloan-Kettering. Private funding kept immunotherapy research alive when the mainstream consensus blocked government grants. New kinds of financiers that specialize in backing unconventional ideas, such as professionally managed venture-capital partnerships and informal networks of angel investors, have increased the potency of high-tech entrepreneurs. The upstart Apple of the 1970s and 1980s relied on semiconductors developed by large chipmakers. Now, developers of mobile phone apps depend on the app stores and secure order taking processes maintained by today's behemoths, Apple and Google.

Problems, Antidotes and Tangles. Massively multiplayer innovation also poses problems. For instance, widespread venturesome consumption supports extensive specialization of development and high-volume production, but it also increases the difficulties of identifying and satisfying buyers' wants. Developers cannot easily anticipate what combination of features will best attract dispersed customers. In preindustrial times, buyers could directly tell artisans what they wanted. Even Edison, who did not customize products but targeted a niche of wealthy patrons, faced less difficulty.

Similarly, global supply chains make products affordable but are harder to coordinate than low-volume production within a self-contained workshop. A geographically dispersed, multilingual workforce makes communication more difficult, precisely codified targets and schedules more necessary, and carefully delegating responsibilities and authority crucial.

Many hands don't always lighten development work, and too many cooks can spoil the broth. As Frederick Brooks wrote in his celebrated book on software development, "The Mythical Man-Month: Essays on Software Engineering": "When a task cannot be partitioned because of sequential constraints, the application of more effort has no effect on the schedule. The bearing of a child takes nine months, no matter how many women are assigned."¹⁴ In fact, 'Brooks's Law' suggests that increasing the size of software teams may delay development. Likewise, conflicts within multi-disciplinary teams can produce deadlocks and clumsy compromises – the proverbial camel crafted by a committee formed to design a horse.

These problems have spurred the development of managerial techniques to support multiplayer innovation. Market research and advertising techniques can help innovators design and market products for mass consumption. Nineteenth-century inventors like Thomas Edison designed and marketed expensive novelties in a more improvised way. Similarly, contemporary organizations have routinized processes to make the diverse expertise of 'many heads better than one.'

Yet, proliferating techniques can magnify the anxieties of overwhelming choice. Some techniques may appear more 'scientific' and thus more appealing. But, methods for discovering and validating general

scientific principles are often ill-suited for developing new products (as discussed in the Appendix). For example, using randomized controlled tests in the early stages of product development can force designers to consider fewer alternatives than more ad-hoc and judgmental rapid prototyping techniques.

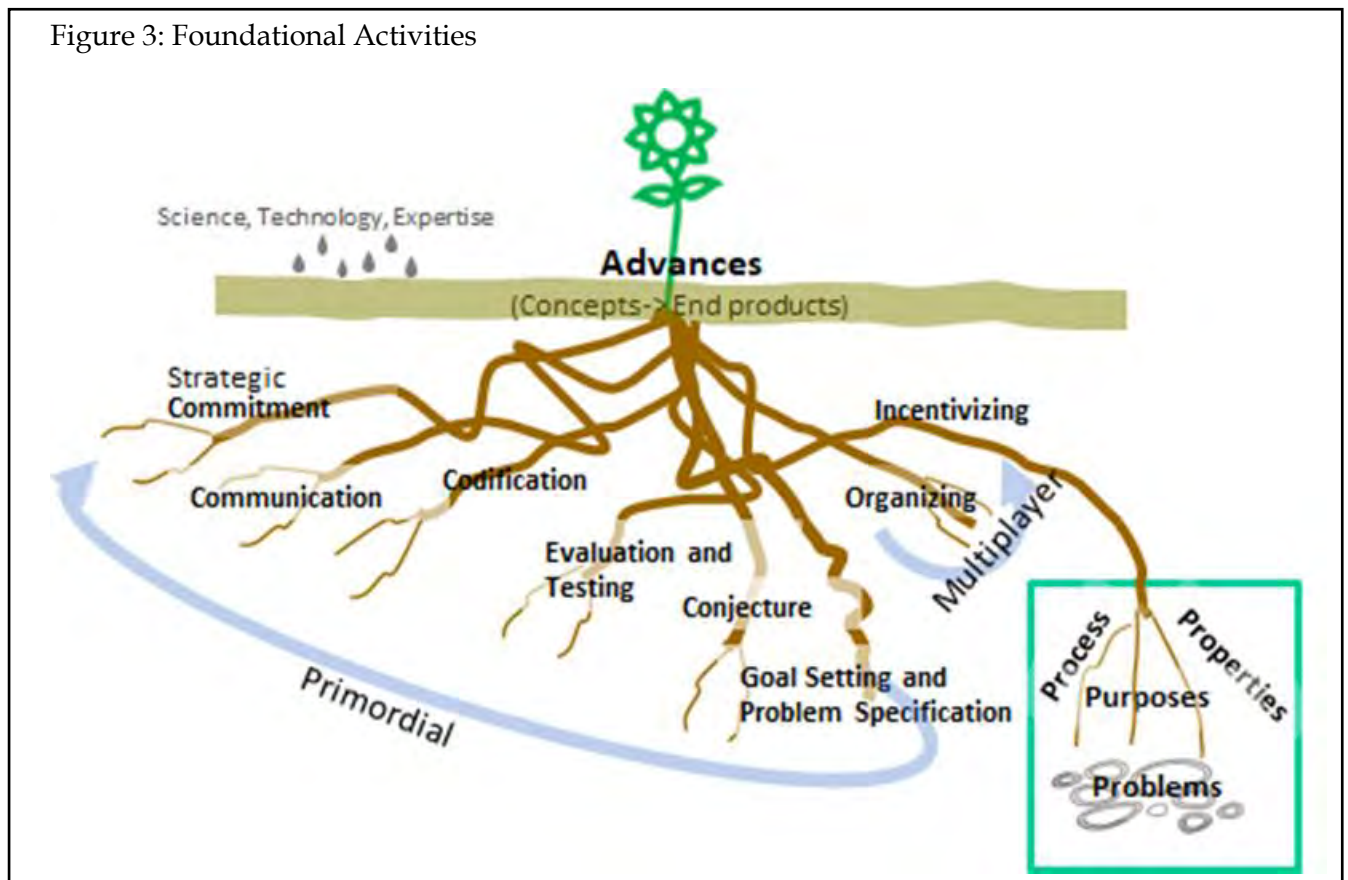
In fact, systematized techniques of any sort have limitations. Codified routines are typically slower than hunches, informal consultations, or unspoken conventions and can suppress the venturesome spirit of innovation. Ritualized market research can discourage imaginative leaps, and by-the-numbers evaluations of new product sales can prevent organizations from persisting with visionary initiatives. Checklists can avoid mistakes in complex operations, but their one-size-fits-all construction can prevent rapid adaptation to local circumstances. Moreover, techniques tend to become increasingly complicated for good and bad reasons. And, as the thicket of complicated techniques grows, so can their ritualistic, deadening use.

A Wikipedia or detailed user manual won't lead us out of these tangles. Nor can we expect a rigorous technique to choose techniques. But a simple guide (or "walking stick"¹⁵) would be helpful.

III. Simplifying Framework

An Octet of Activities. My simplification highlights eight foundational or "root" activities, characterized by their "function," or the kind of "effect" they produce. These are depicted in Figure 3, a more detailed version of Figure 2.

Figure 3: Foundational Activities



Six of the eight activities, namely **goal and problem specification**, **conjecture**, **evaluation and testing**, **codification**, **communication**, and **strategic commitment**, serve ‘primordial’ functions: they have long been necessary for significant advances, as suggested in the first section of this Note. The other two – **organizing** the specialization of innovative contributions and **incentivizing** contributors¹⁶ – reflect the distinctive problems of multiplayer innovation. As mentioned, transformative innovations now combine numerous specialized contributions. Notwithstanding the faith of some in the coordinative powers of market prices, effective alignment requires designing and implementing organizational structures, routines, and incentives. These activities are at the heart of modern managerial work.¹⁷

Like underground roots that support the more visible growth of shoots to fully formed plants or trees, the activities, often undertaken behind the scenes, support a more visible progression from novel concepts to useable innovations. Thus, the *goal* of early cancer diagnosis leads to a *conjecture* about magnetic resonance imaging (MRI) devices. Repeated *evaluation and testing* turns the concept into a *codified* design implemented by medical product companies *committed* to investing in diagnostic devices by *organizing* multifunctional teams that include *incentivized* salespeople who *communicate* with doctors and hospitals. And like roots that grow along with the plants they nourish, activities evolve with the innovations they support. For example, organizing and incentivizing a sales force becomes more salient after a novel device or drug concept becomes a saleable product.

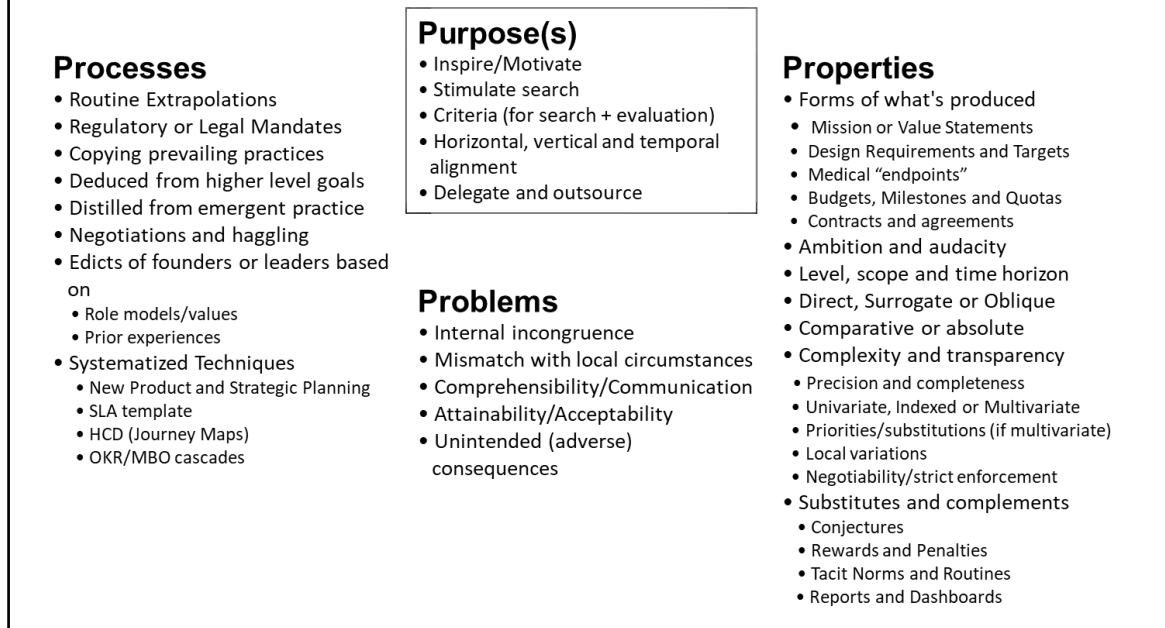
Like roots which absorb and conduct water and dissolved minerals from remote watersheds – and oxygen from immediate soil – innovative activities draw on available science, technology, and expertise. For example, MRI devices were based on the scientific discovery that magnetic fields produce distinctive resonances in the nuclei of different atoms. Yet, unlike actual roots, activities do not passively or mechanistically absorb and conduct science, technology, and expertise. Imaginative humans creatively select, combine, and adapt, what’s available. Their creative use of existing ideas can in turn produce new ideas and innovations that other innovators can use. The development of versatile antibiotics supported chemotherapeutic treatments of cancer. Innovative activities thus enhance the watersheds they draw from.

The simplification does not cover all possible activities. Rather it reflects my judgment about multiplayer innovation in the broadest possible range of domains. Likewise, it does not delineate or demarcate sharp differences. Like actual roots that anchor, nourish, and sometimes store food, activities can be multi-functional, producing more than one desired effect. For example, besides evaluating the effectiveness of a drug, a clinical trial can suggest conjectures about additional uses. Planning meetings can align overall goals, codify specific targets, and organize who does what. Similarly (and like actual roots), activities overlap and crisscross. Codified goal setting can provide metrics for evaluation and testing and for incentivizing effort.

For each activity, we can further examine (as also shown in the Figure 3 inset) its:

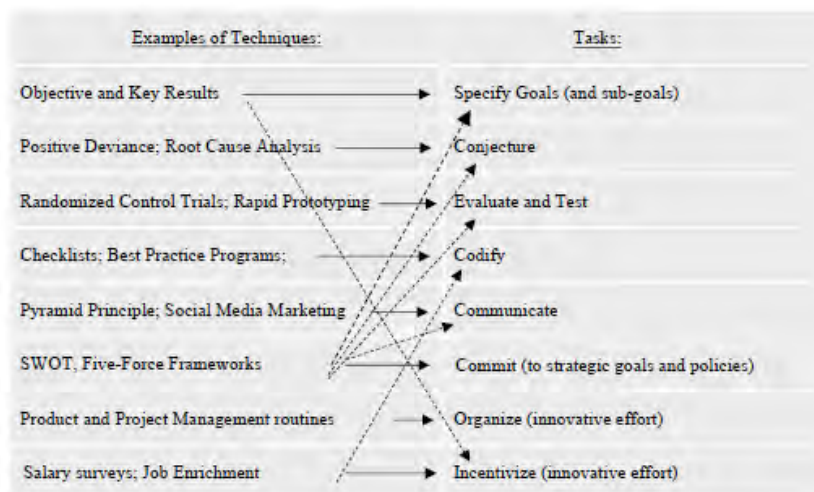
- *Purposes* – the more precise “whys” for undertaking an activity (or the specific effect intended).
- *Properties* – the form and attributes of “what’s” produced (to secure the desired effects.)
- *Processes* – “how” activities are performed (including, but not limited to, the systematic techniques and procedures used).
- *Problems* – what might stop achieving the intended purpose(s) or produced undesired effects.

Figure 4: Attributes of Goal Setting Activities



Each of the four attributes spans several possibilities (as indicated for 'goal setting and problem specification' in Figure 4). The possibilities again interact and overlap. Intended purposes, for example, can affect the problems that arise. Similarly, like algebra and arithmetic, managerial techniques can support many activities. For instance, Human Centered Design techniques help innovators specify goals for new products, develop creative conjectures, and test these conjectures. (See Figure 5 for other examples).

Figure 5. Examples of Versatile Techniques



Because of multi-functional activities and techniques, my framework isn't "mutually exclusive and collectively exhaustive." It isn't like a biological classification, which assigns organisms to a single category. Rather, it offers labels for tagging common activities and techniques. Just as "anchoring," "absorption" and "transmission" are used to describe the functions of roots – or, in our earlier analogy, National Geographic guides depict "campsites" and "recreational areas" – the labels help organize productive knowledge. Moreover, the organization of productive knowledge is itself designed for practical uses such as:

Diagnosing Problems. Like a guide map, the framework can help address "Are we on the right track?" and "What problems lie ahead?" questions. Such assessments might include asking whether the different activities are being performed consistently, are congruent with external circumstances, and are likely to produce the desired results. The framework can similarly help diagnose misalignments. For example, an analysis of goal setting activities may show that capricious or inflexible processes may undermine the motivational purpose of goal setting.

The framework can similarly help identify outdated practices. The rapid growth of knowledge creates a gap between what's 'out there' and what individuals and organizations use or even know about. Mature organizations are particularly likely to fall behind. New organizations start with a clean slate and have the incentive to pick the best techniques they can afford. But success produces complacency and complexity that make change difficult. Recruiting entry-level employees who have just graduated from schools and colleges (whose curricula also often lag) and promoting from within, reinforces insularity. Many new techniques (such as Human Centered Design) cross standard organizational boxes. No one, therefore, has the responsibility or authority to learn or apply the new knowledge.

A review of how organizations undertake the broadly defined activities of my framework can alert them to gaps and blind spots. Moreover, because the framework focuses on widely undertaken foundational activities, it permits looking for techniques from far and wide.

Systematic reviews do not, however, eliminate the need for judgment. As mentioned, goal setting can be characterized by many purposes, processes, properties, and problems. The possibilities for misalignments and obsolescence are vast, and some issues are better tolerated than treated. For instance, while mechanically codified targets can make large organizations inflexible, new businesses can implode if they don't systematize goal setting. In other words, finding an apparent misalignment should prompt the question of "what's going on here" – and why – rather than immediate changes.

Adapting Popular Techniques ("Best Practices"). Copying even rudimentary products and procedures requires knowing what to replicate. The proverbial wheel invented circa 3,500 B.C., needed axles with smooth and round ends that fit snugly into wheel holes while leaving room for the wheels to rotate. The insides of wheel holes also had to be smooth and round. Wheels could, therefore, not be replicated just by seeing a cart roll by. Similarly, reusing complex multiplayer practices requires selective

imitation and careful adaptation (See Box ‘Modernizing Japan’). My framework provides a starting point for choosing what to copy exactly, as well as what to adapt and how.

Modernizing Japan

After extended isolation that led to humiliating military defeats in the mid-19th century, Japanese officials made an all-out effort to learn from the West. In 1872, the Iwakura Mission traveled around the world, touring factories and studying legal systems and social customs. French experts were hired to help draft a new legal code, British experts provided advice on industry, and Americans gave guidance on agriculture and education. Prussia provided a model for the army. Diplomats started to dress in coats and tails instead of kimonos. The Emperor could be seen wearing military uniforms, and the Empress in Victorian gowns. But Westernization wasn’t blind. Unlike Türkiye after Atatürk, Japan did not adopt the Roman script. A new “*bunmei kaika*” (“civilization and enlightenment”) policy did not grant Japanese women the personal freedoms that members of the Iwakura Mission had been surprised to find women enjoyed in the United States.¹⁸ And a new wardrobe did not alter the Emperor’s divine status.

Learning from Case Histories. Analyzing cases can complement studying techniques in several ways. For instance, the heuristic of “loose-tight” controls¹⁹ can guide organizations seeking a middle ground between comprehensive top-down planning and uncoordinated individual initiative. Studying specific cases can help us learn what warrants tighter or looser oversight and control, even if the cases themselves do not make this explicit. Similarly, cases can also get us to think about “sweet spots” for the applicability of techniques.

How studying specific cases improves the utility of techniques and vice versa is comparable to the symbiotic benefits of studying great novels and writing conventions. Aspiring writers may learn more about plot and character development from great novels than from studying the conventions of writing. However, knowing the conventions increases what aspiring writers learn by guiding their attention to how a great novel develops plot and character or deviates from standard conventions.

Specific cases also help adapt and cross-fertilize exemplary systems and architectures. As mentioned, practical designs embody many choices. But no one can anticipate the right constellation of choices from a nearly infinite number of possible combinations. Instead, pioneering developers make a guess about their initial designs, which they and others subsequently modify, again through guesswork. This can take decades. Studying the architecture of successful designs facilitates adaptation and replication of the hard-won advances. Knowing *how* the elements of a complex system work together and align with exogenous circumstances provides valuable hints about what might need to be changed.

The histories of transformative advances can tell us even more about their whys and wherefores. For example, Marvin Bower, who founded McKinsey & Co. and practically invented management consulting, used prestigious law firms and the legal profession as his base model. But Bower did not copy blindly. He took what he expected would fit and invented the rest. With experience, Bower and his partners then gradually modified their model. Studying [how the McKinsey system evolved in its first decades](#) can provide insights into the logic of its basic architecture (which is not self-evident today) and its applicability to other domains.

Case histories also address the problem described in G. K. Chesterton's thought experiment of "a fence or gate erected across a road. The more modern type of reformer goes gaily up to it and says, "I don't see the use of this; let us clear it away." To which the more intelligent type of reformer will do well to answer: "If you don't see the use of it, I certainly won't let you clear it away. Go away and think. Then, when you can come back and tell me that you do see the use of it, I may allow you to destroy it." Historical facts provide more insights about the unintended consequences of removing Chesterton fences than just the "go[ing] away and think[ing]" that Chesterton suggests.

At the same time, without some basis for thoughtful interpretation, potentially important historical facts may be ignored. Moreover, an overwhelming number of facts and interrelationships make learning from the evolution of complex systems difficult. My labels provide a starting point for grouping facts and then looking for relationships between a manageable number of groups.

Finally, case histories can show the vital contributions of human imagination, persuasiveness, occasional ruthlessness, and fortitude. Transformative advances require broad-based movements with visionary leadership. Success may bring great financial rewards, exhilaration, and possibly a place in history, but visionaries also risk ruinous loss, frustration, and obscurity. To take such a path requires a love for challenging accomplishment. Paradoxically, carefully analyzing the difficulties can stimulate the pursuit of perilous adventure.

Appendix: Scientific vs Technical Knowledge ('Technology')

Interdependencies and Similarities.

Scientific discoveries often play a pivotal role in developing new products and technologies. Thus, the discovery of nuclear magnetic resonance prompted the development of industrial spectrometers used to analyze the composition of chemicals. In some instances, scientific understanding that came after the development of new products has helped improve the products: thermodynamics improved the efficiency of steam engines, for example.²⁰ Bacteriology and virology have improved the development of vaccines (which Jenner pioneered in Britain before scientists showed how bacteria and viruses cause disease). And practical problems can prompt scientific research that helps solve the problems. Famously, Pasteur identified microorganisms responsible for fermentation after brewers asked for help in limiting spoilage. Stokes, therefore, calls scientific research directed to practical ends "Pasteur's Quadrant" research.²¹

Conversely, new technologies can advance scientific understanding. Recounting Henderson's quip that "until 1850, the steam engine did more for science than science did for the steam engine," physicist Malcolm Longair writes that James Watt's 1765 invention of a condenser, made in the course of repairing a steam engine, "led to the underpinning of the whole of thermodynamics."²² Similarly, the invention of electron microscopes brought to scientists' attention naturally occurring phenomena they could not otherwise observe. Additionally, new instruments such as spectrometers enabled the testing of scientific theories. Going back much further to the Enlightenment, clocks produced for human needs helped inspire scientific efforts to debunk animistic theories (that, for instance, gave stones the 'will' to fall).²³ The development of telescopes provided the basis for Galileo's -and later Newton's- revolutionary theories of planetary motion.

Similar human qualities, values, and skills drive both scientific research and new technologies and products. Unlike biological evolution, both are propelled by human striving and creativity and not just by chance. Both seek to learn from mistakes and extend prior successes. Both can require persistence – the discovery of the structure of DNA and evidence of the existence of Higgs boson ("God") particles no less than the 19th-century dream of controlled fixed-wing flight. Both also value observable phenomena. And both require tacit skills that cannot be precisely codified.

Scientific research and technological development combine increasingly dispersed contributions. Enlightenment scientists, like clockmakers and shipbuilders, pooled ideas from countries that periodically fought bloody wars. Today, scientific research, like commerce and industry, draws on contributions from every continent. And like the massively multiplayer development of useful products, globally dispersed scientific research requires aligning scientists' goals, conjectures, tests, codification, and communication.

Obvious and Subtle Differences.

While science may spur the development of spectrometers, improve steam engines, and help brewers, it cannot provide all – or even the greater part – of the knowledge that manmade products embody. This is an intrinsic feature of scientific knowledge, not a defect. In his seminal work, *The Scope and Method of Political Economy* (1890), John Neville Keynes (father of John Maynard) argued against confusing the science of economics with value-laden concerns about economic ends. Keynes also distinguished economic science

from systematic techniques for attaining desired ends. Arguably, the distinctions help produce more and better science; if you want to find the glacial headstreams of a river, stay away from the tributaries in the plains.

But, as mentioned in the main text and shown in Figure 1, the knowledge required to develop new products includes codified technologies, and tacit and contextual expertise. To rephrase Schumpeter: apply as much electromagnetic theory as you please, you will never get a maglev train thereby. Similarly, the social sciences may offer general directions and signposts but cannot by themselves supply the organizational techniques that undergird multiplayer innovation. Just applying cutting-edge economics, sociology, or psychology could not have produced Intel's goal-setting system. Moreover, methods designed for scientific discovery are often ill-suited for choosing "upstream" ends, developing systematic "midstream" technologies, or acquiring tacit "downstream" expertise.²⁴

Although scientists often use specific facts to form conjectures and tacit knowledge to perform experiments, science is self-evidently different. Scientists themselves often emphasize the distinctions between objective scientific propositions and value-laden choices of ends that fall outside the scope of "positive" scientific research.*

The differences between science and systematic techniques are more subtle, particularly in technical fields such as engineering that draw heavily from scientific propositions. Yet, as Stanford engineering professor Walter Vincenti observes in *What Engineers Know*, "technology, though it may apply science, is not the same as or entirely applied science." Rather, it is "an autonomous body of knowledge, identifiably different from the scientific knowledge with which it interacts" (See Box 'Walter Vincenti: What Engineers Know'). To make Vincenti's point more colorful, the Mona Lisa is more than just applied paint, although Leonardo did apply paint to produce his masterpiece.

Walter Vincenti: What Engineers Know

"Modern engineers are seen as taking over their knowledge from scientists and, by some occasionally dramatic but probably intellectually uninteresting process, using this knowledge to fashion material artifacts. From this point of view, studying the epistemology of science should automatically subsume the knowledge content of engineering. Engineers know from experience that this view is untrue... my career as a research engineer and teacher has been spent producing and organizing knowledge that scientists for the most part do not address."²⁵

Science has become increasingly important to engineering since Vincenti wrote his landmark book more than three decades ago. Conversely, increasingly sophisticated instruments have increased scientists' reliance on engineering. Nonetheless, significant differences remain between engineering and scientific knowledge – and between other kinds of systematic technical knowledge (of fields such as medicine) and the science they use. How the knowledge is produced – how scientific communities and developers of

* Economists and other researchers who seek to understand social phenomena in a scientific way may include norms and objective functions in their theories, but primarily to better explain the consequences of people's normative choices or the reasoning behind them. Like natural scientists, social scientists also exclude ethical inquiry about what people should want.

systematic techniques set goals, form conjectures, codify and communicate results, and so on – is also different. In fact, the increased dispersion and specialization of scientific research and technological development may have widened the gap, as we will see.

Internal Consumption and Production of Science.

Modern scientific communities are highly self-contained and autonomous; they produce knowledge mainly for internal use. Scientific knowledge may also have value in products used by non-scientists, but that is not a necessary purpose. For many decades, the existence of the Higgs field was regarded as the central problem in particle physics, even though it had no apparent practical consequence.

Even scientific research in “Pasteur’s quadrant,” which is prompted by practical problems, is generally insulated from developing its practical applications. Discovering the pathogen causing AIDS had practical urgency: it would provide the basis for a diagnostic test. But while the design of the test kits had to consider practical issues of large-scale production, distribution, storage, usability, regulatory compliance, etc., the scientific hunt for the pathogen could be insulated from that endeavor. Furthermore, the “worth” of scientific results can transcend their direct utility. A scientific discovery that does not provide a direct or obvious solution to the practical problem it was funded to solve may nonetheless be celebrated as a valuable advance. Linus Pauling and his colleagues demonstrated in 1949 that sickle-cell disease occurs because of an abnormality in the hemoglobin molecule. Although the disease remained incurable for decades, this discovery has been judged as a milestone in the history of molecular biology.

The communities that produce scientific research also specify questions that merit investigation, the range of hypotheses advanced, and the kind of reasoning and evidence they consider legitimate. Particle physicists set the standards for evidence that would demonstrate the existence of the Higgs field. Fellow virologists evaluated the research produced by virologists at the Pasteur Institute in France and the National Cancer Institutes in the U.S. that identified the retrovirus now known as HIV-1 as the cause of AIDS. Even when scientists seek outside funding for scientific research with an explicit practical end, funding agencies turn to the scientists’ peers to evaluate the research proposal.

Internal use and evaluation significantly simplify coordinating scientific research. Even if the members of a community are geographically scattered, similar training and backgrounds make them epistemically close. Scientists can, therefore, relatively easily anticipate what will appeal to ‘buyers’ and how to ‘sell’ their work. Even when scientists do their research to help develop products (such as an HIV test), they usually don’t need to consult end users. And, to the extent their research is self-contained, scientists don’t have to coordinate with individuals and groups outside their community.

Internal evaluation also allows – although it doesn’t require – scientific communities to privilege, as Thomas Kuhn termed it, “paradigmatic” ideas. Simply put, we can think of these ideas as the core assumptions that the members of a community take for granted and which bound the hypotheses they propose and test.²⁶ The paradigmatic ideas – in conjunction with the norm of citing and building on prior work – align the research of competing individuals and groups who are also expected to make novel and creative contributions and facilitate the efficient communication of the results. Moreover, as scientific communities become more globally dispersed, paradigms are vital in limiting fragmentation and balkanization.

This is not to suggest that paradigms require scientists to forever march along the same path. As Kuhn pointed out, the accumulation of anomalies can precipitate a revolutionary collapse of paradigms. Scientists can also drift away from the conjectures and questions framed by their community's paradigm. But, in either case, paradigms typically continue to align scientists' assumptions and hypotheses, either because a new paradigm follows a revolutionary collapse or scientists who drift away from the mainstream create a new community with a new paradigm.

Paradigmatic Conjectures and Tests.

Although the paradigms of different scientific communities encourage them to research different kinds of questions, they will generally tend to favor hypotheses (or what I have called "conjectures") that are:

- *Precisely, and preferably concisely, codified* — Newton's second law of motion, $F = ma$, and Einstein's law of mass-energy equivalence, $E = mc^2$ provide ideal examples.
- *Universal and timeless* — propositions are treated as scientific to the extent they abstract away from specific circumstances of place and time. Even in common usage, the more general a proposition, the more "scientific" it is considered.²⁷
- *Objectively verifiable* — through decisive tests that satisfy fellow scientists

Preferences for precise specification, universality, and verifiability reinforce each other. For instance, scientists cannot verify imprecisely formulated hypotheses. Similarly, scientists tend to avoid events that occur in a particular time and place because many plausible but unverifiable 'just-so stories' can be told about the causes. And, like the specific paradigms of individual scientific communities, the general preferences of scientists also promote cohesion and reduce the need for managerial techniques to pool dispersed individual effort. For instance, precise specifications and standardized verification allow scientists to communicate with each other efficiently and to rely on each other's work (without everyone replicating each other's results).

The degree to which different scientific communities require precise specification, universality, and objective verification varies (See Box 'Variations in Conjectures and Tests'). But even that aspiration, widespread in science, is not a common feature in developing new technologies and products, as we will see.

Variations in Scientific Conjectures and Tests.

Not all scientific knowledge is concise – as anyone who has had to memorize the periodic table will testify – and cell biologists, ecologists, and zoologists treat detailed descriptions as contributions. But, scientific communities that start with sprawling collections of facts strive for concise propositions. Science advances with "general statements of steadily increasing explanatory power," according to zoologist Peter Medawar. These statements "annihilate" the need to know particular facts. "Biology before Darwin was almost all facts," writes Medawar, but now is "over the hump."²⁸ Generality also seems to affect status. August Comte, considered the first modern philosopher of science, arranged the sciences "in the order of generality of the principles they establish[ed]."²⁹ Molecular biologist James Watson dismissed naturalist colleagues at Harvard who engaged in classification as "stamp collectors."³⁰

Similarly, paleontologists do research and inconclusively argue about the one-off extinction of dinosaurs. But even in these instances, scientists reject evidence that lies in a particular beholder's eye and strive to develop more conclusive tests. As the evolutionary biologist Jonathan Losos puts it, his field was thought to be like history for the first century of its existence: "You can't go back in time and see what happened, so you just have to try to figure it out." Now, researchers "replay the tape" using microorganisms to evaluate hypotheses in their laboratories.³¹

Enforcing Scientific Standards.

Scientific communities face strong incentives to enforce their paradigmatic norms strictly. While science may be highly "epistemically" self-sufficient, scientists require outside funding. But, governments, foundations, and philanthropists who provide the funds cannot, as mentioned, independently assess the quality of the research. Instead, the outside funding agencies rely on certification provided by journals, whose referees and editors enforce rigorous adherence to the research community's standards for parsimony, precision, and testing. Similarly, not tolerating mistakes helps scientific communities and publications avoid externally damaging perceptions of favoritism. Therefore, if referees raise credible objections, scientific papers aren't accepted for publication. And increased competition between communities for outside resources and standing has likely spurred a tightening of criteria for hypotheses and evidence and reduced the scope for deviant or idiosyncratic inquiry. It also increases scientists' confidence in each other's work without requiring deep knowledge of individual researchers who, as mentioned, are now widely geographically dispersed.

Along with—and possibly because of—stricter criteria, scientific communities have increased qualifications for membership. Bodies such as the Royal Society once included well-born gentleman scholars and even the Delft cloth merchant Antonie van Leeuwenhoek, now considered the Father of Microbiology. Today, individuals who do not have PhDs and jobs at universities or recognized research institutions have been almost completely marginalized. Concurrently, the number of research communities and the specialization of its members have also grown. Thus, while more opportunities for higher education and the public funding of scientific research have made entering scientific communities more meritocratic and open to the not-so-well-born, credentialed specialization has limited membership of specialized communities to individuals with similar knowledge, training, and career experiences.

Contrast with Technological Standards.

Outside Evaluation. In contrast to scientific research, outside users have a crucial role in assessing new products and, thus, implicitly, the technologies the products embody. Visionaries may develop products far ahead of anyone's articulated wants, but ultimately, their success requires buyers to open their wallets. This does not mean that users always know what's good for them – patients continued to demand bloodletting from their sometimes-reluctant physicians through the mid-18th century. Even today, patients will ask for tests and treatments that doctors discourage. Moreover, the preferences of outside buyers are less predictable than the internal paradigmatic preferences of scientific communities. Buyers won't say what they want except for customized goods like kitchen cabinets. Yet their wants are often inchoate and fickle. Today's venturesome consumers are not merely willing to take their chances on novelties; they frequently demand surprising, unexpected features.

Incomplete Codification and Contextual Dependencies. Developers and users of products cannot depend just on parsimonious, precisely specified knowledge favored by scientists. As mentioned, manmade products embody a complex tangle of knowledge. Some of this knowledge is precisely specified – in engineering drawings, circuit diagrams, and project plans. But knowledge used to design, produce, and use new things also inevitably has tacit complements. Precise specification can also be expensive or even harmful. For instance, it may be cheaper to let employees learn by doing and more effective to allow them to adapt to changing circumstances than to precisely specify (a la Henry Ford) how they should perform assigned tasks.

Generalizability across uses and time has similar tradeoffs. All airplane designs must conform to universal laws of nature; but there is value in adapting designs to intended use (e.g., long-haul versus short-hop, or cargo versus passenger). Moreover, given the practical difficulty of getting something to work, developers will often first tune their designs for specific circumstances and specific users and then look for ways to generalize their designs for broader applications.

Useful products also must match changes in tastes and the Zeitgeist. Unlike scientists who seek to discover the unchanging laws of nature, product developers cannot produce timeless designs. And besides changing tastes, increasing use itself can affect utility. For instance, the capacity of standardized credit scoring to predict loan defaults deteriorated when its increased use by lenders taught borrowers how to game their scores. Similarly, the wide use of antibiotics helps breed resistant pathogens. Conversely, learning or network effects can increase utility. For instance, the popularity of a surgical procedure can accelerate its improvement, and the wide adoption of a programming language such as Java can make it a valuable standard. In contrast, increased acceptance of a scientific hypothesis does not affect its correspondence to nature. Whatever reality is “out there” remains unchanged.

Flexibility of Standards. Developers of new products and technologies face less rigorous standards than those imposed by gatekeepers of scientific research because users consider mainly their own costs and benefits (rather than enforce a group norm). Thus, unlike referees of journal articles, users of new products are often willing to tolerate obvious limitations in the expectation that they will be fixed. In some cases, the expectation can even lead to acquisitions of buggy “first generation” products that make users temporarily worse off.³²

Membership criteria for joining the multiplayer innovation game are also more flexible than the criteria now imposed by scientific communities. As in the sciences, the increased division and specialization of labor in developing and using new products has raised standards for the qualifications required of many specialists. However, there are significant differences. New technology and products have continued to provide entrepreneurial opportunities for college dropouts like Bill Gates, Steve Jobs, and Mark Zuckerberg (who would now be excluded from scientific communities). The companies they have founded (Microsoft, Apple, and Facebook) recruit many self-taught hackers. Moxie Marlinspike, whose encryption programs have been embedded in applications used by billions, barely finished high school before finding a job in Silicon Valley. Moreover, greater inclusivity isn’t the result of more fair-mindedness or impartiality. Rather, global competition to satisfy demanding consumers requires integrating the efforts of a broader range of talents and skills than producing scientific research for specialized communities.

Systematic Techniques. Outside evaluation (and the other distinctive features of the development of new products and technologies) have stimulated the development of techniques (discussed in the main text) not widely used by contemporary scientific communities. For example:

- Anticipating ‘outside’ users’ often inchoate wants (rather than predicting what like-minded scientific colleagues will value) has spurred conjoint analysis, focus group, and ‘design thinking’ techniques. Similarly, while scientists continue to depend mainly on traditional journal articles and conferences to disseminate their findings, developers of commercial products use many new communication techniques such as YouTube videos, tweets, podcasts, and pop-up stores to inform and persuade buyers.

- Evaluating complex products that must satisfy several performance, safety, legal, and societal requirements has spurred computer simulations, rapid prototyping, A/B testing, field observations, and other techniques. Unlike experiments undertaken to satisfy scientific colleagues (and journal referees), these techniques aren’t designed to produce decisive validation of a parsimonious hypothesis; instead, the techniques seek to incorporate all the external factors likely to affect the performance of products under conditions in which the products will be used, instead of “controlling” for these factors.

- Efforts to coordinate diverse activities performed by widely dispersed individuals with different predispositions and training (rather than by communities of like-minded scientists) have spurred new methods for setting goals, motivating employees, structuring organizations, and managing projects that cross organizational boundaries.

Concluding Comments

Like Vincenti, Henry Petroski, Professor of Engineering and Professor of History at Duke University, has emphasized the difference between technology and science in numerous books and articles. In the first paragraph of *The Essential Engineer*, Petroski writes that “both medicine and engineering do use scientific knowledge and methods to solve relevant problems, but neither is simply an applied science. In fact, the practices of medicine and engineering are more like each other than either is like unqualified science.” Yet, Petroski continues, “the word science is commonly understood to include medicine, engineering, and high-technology.”³³

Petroski offers a convenient distinction: “*Science is the study of what is; engineering is the creation of what never was.*” But confusion arises because “even in their most basic professional activity, scientists can act like engineers (and vice versa). Chemists regularly synthesize new compounds, and biologists create new strains of plants and animals that do not exist in nature. In other words, scientists can do engineering (as engineers can do science).”³⁴

While “Science” is a “useful shorthand for a wide range of activities,” Petroski complains that the expansive label also “obscures differences” and gives science “a primacy that it may or may not deserve.” Petroski cites several examples, going back to the 1950s, of newspapers attributing engineering successes to “science” and “scientists” – while attributing failures to “engineering” or “engineers.” This bias could reflect educational backgrounds; Petroski observes that nearly all science reporters study science, not engineering, in college. Or it might come from a “Western Platonic bias” that views “scientists who deal in ideas, even ideas about things... as superior to engineers who deal directly in things.”³⁵

The lower standing “provides fodder for engineers who feel that their profession is misunderstood and undervalued.” Some see a reflection of the “hierarchical structure between the sciences and engineering” at prestigious research institutions in inaccurate media accounts rather than “just innocent confusion or

carelessness.”³⁶ But, whatever the cause, confusing science and engineering “can leave politicians, policymakers, and the general public unable to make informed decisions,” including decisions about allocating research funds.³⁷

This appendix (and the main text) suggests there is even more at stake than status or resources from the misapplication of scientific sensibilities and methods in many practical fields beyond engineering.

References

- Baron J. H. (2009). Sailors' scurvy before and after James Lind—a reassessment. *Nutrition reviews*, 67(6), 315–332. <https://doi.org/10.1111/j.1753-4887.2009.00205.x>
- Bhidé, A. (2008). *The venturesome economy: How innovation sustains prosperity in a more connected world*. Princeton University Press.
- Bhidé, A. (2017). Constraining Knowledge: Traditions and Rules that Limit Medical Innovation. *Critical Review*, 29(1), Article 1. <https://doi.org/10.1080/08913811.2017.1288464>
- Brooks, F. P. (1975). *The Mythical Man-Month: Essays on software engineering*. Addison-Wesley Pub. Co.
- Elster, J. (1993). *Explaining technical change: A case study in the philosophy of science*. Cambridge Univ. Press.
- Hayek, F. A. (1945). The Use of Knowledge in Society. *The American Economic Review*, 35(4), 519–530.
- James, W. (1907). *Pragmatism: A new name for some old ways of thinking : popular lectures on philosophy* (1st ed.). Longmans, Green. <https://hdl.handle.net/2027/hvd.32044009965229>
- Knight, F. H. (1921). *Risk, Uncertainty and Profit*. Houghton Mifflin Company.
- Longair, M. S. (2003). *Theoretical concepts in physics: An alternative view of theoretical reasoning in physics* (Second edition.). University Press.
- Medawar, P. B. (1984). *Pluto's Republic*. Oxford University Press.
- Peters, T. J., & Waterman, R. H. (1982). *In search of excellence: Lessons from America's best-run companies* (1st ed). Harper & Row.
- Petroski, H. (2010). *The essential engineer: Why science alone will not solve our global problems* (1st ed). Alfred A. Knopf.
- Roethlisberger, F. J. (Fritz J. (1977). *The elusive phenomena: An autobiographical account of my work in the field of organizational behavior at the Harvard Business School*. Division of Research, Graduate School of Business Administration, Harvard University.
- Rosenberg, N. (1976). *Perspectives on technology*. Cambridge University Press.
- Shapin, S. (1998). *The scientific revolution* (Paperback). University of Chicago Press.
- Stokes, D. E. (1997). *Pasteur's quadrant: Basic science and technological innovation*. Brookings Institution Press.
- Usselman, S. W. (1992). From Novelty to Utility: George Westinghouse and the Business of Innovation during the Age of Edison. *Business History Review*, 66(2), 251–304. <https://doi.org/10.2307/3116939>
- Vincenti, W. G. (1993). *What Engineers Know and How They Know it: Analytical Studies from Aeronautical History*. Johns Hopkins University Press.

Notes

¹ We often accept or reject options just in our minds. We don't expose every possibility that we might think of to a competitive battle for survival outside our imaginations and even our external evaluations reflect our choices of test designs and interpretation of the results.

² Willful humans can dismiss favorable options – or even accept unfavorable options – “in order to gain access to even more favorable ones later on.” (Elster, J. 1993. *Explaining technical change: A case study in the philosophy of science* Cambridge Univ. Press p. 91). In contrast, natural selection has an “impatient, myopic, or opportunistic” character. It cannot learn from mistakes because it has “no memory of the past,” and no forethought – it does not forgo favorable mutations now to realize better ones later, as it has “no ability to act in terms of the future.” (Elster 1993 p. 51) Moreover, nature does not permit willful imitation: house cats cannot follow the hunting habits of tigers. Mutations diffuse entirely as a side-effect of reproduction.

³ As the Chicago economist Frank Knight wrote in *Risk, Uncertainty and Profit* (1921)

⁴ *Contra* Schumpeter's “gales of creative destruction” imagery however, the alternative technologies can take decades to gather force.

⁵ And possibly the existential anxiety that Kierkegaard said attends such leaps.

⁶ According to Dewey, even the most thorough and careful inquiry could at best produce “warrantable assertions” – provisional, more-or-less reliable claims, supported by a reasonable warrant.

⁷ James 1907 p. 22

⁸ John Dewey devoted his life to radically reforming education, while James suggested unusual measures to increase one's productive working hours by curtailing sleep. Later 20th century “neo-pragmatist” philosopher Richard Rorty promoted *Social Hope* (for a “global, cosmopolitan, egalitarian, classless, casteless society” as he put it in the preface).

⁹ Rosenberg (1976) p.75-76

¹⁰ Usselman (1992) p. 254

¹¹ See Bhidé (2008, chapter 11). Venturesome consumption has not widened uniformly, however. As I have argued (Bhidé 2016) long-standing traditions and contemporary rules have held back medical advances by limiting the role of consumers.

¹² <https://runningtortoiseandhare.wordpress.com/running-shoes/history-of-running-shoes/>

¹³ Feet naturally roll inwards (“pronate”) when they land while running or walking, but excessive or too little pronation can cause injuries.

¹⁴ Brooks 1975 p. 17

¹⁵ To borrow a term from Roethlisberger (1977)

¹⁶ Besides an organization's employees, contributors include financiers, suppliers, and consumers who risk their money, time, and effort.

¹⁷ The managerial activities play an especially important role in initiatives undertaken by large hierarchies (such FAANG and large pharma companies) and networks (such as the National Institutes of Health and ‘open system’ software developers).

¹⁸ Ethan Segal, "Meiji and Taishō Japan: An Introductory Essay" downloaded on August 26, 2018, from <https://www.colorado.edu/cas/tea/becoming-modern/1-meiji.html>

¹⁹ Peters & Waterman 1982.

²⁰ Scientific knowledge can also help control dysfunctional practices – for instance, ignorance that Vitamin C rather than all sour tasting substances prevent scurvy is said to have led to its resurgence when the British Navy substituted lime juice for lemon juice in sailor's diets (Barron 2009).

²¹ Stokes 1997.

²² Longair, Malcolm S. 2003. P. 223. *Theoretical Concepts in Physics: An Alternative View of Theoretical Reasoning in Physics*. Cambridge: Cambridge University Press

²³ Shapin, Steven 1998. *The Scientific Revolution* Chicago: The University of Chicago Press. (paperback edition) p. 32-37

²⁴ An inflexible adherence to methods demanded by scientific communities can therefore hinder the development of useful products. Rather, effective development requires an eclectic choice of techniques – and the willingness to deviate from standards required for peer-reviewed scientific publication.

²⁵ Vincenti 1993 p. 3

²⁶ Scholarly communities in the humanities who have as much autonomy as scientific communities to choose their norms have apparently not favored internal paradigmatic consensus. This may derive from a tradition of contention that preceded the Scientific Revolution. In the sciences, the founding figures, Shapin's (1998) account suggests, explicitly rejected norms of irreconcilable contention.

²⁷ For instance, Hayek (1945) contrasts scientific knowledge of "general rules" with "knowledge of the particular circumstances of time and place."

²⁸ Medawar 1984 p. 29

²⁹ Knight 1921, p. 8

³⁰ Watson may have borrowed his putdown from the physicist Ernest Rutherford who supposedly once said: All science is either physics or stamp collecting. Petroski 2010. p. 33

³¹ Interview with Losos published in Harvard Gazette downloaded from [https://news.harvard.edu/gazette/story/2017/10/evolution-book/?utm_source=SilverpopMailing&utm_medium=email&utm_campaign=10.03.2017%20\(1\)>](https://news.harvard.edu/gazette/story/2017/10/evolution-book/?utm_source=SilverpopMailing&utm_medium=email&utm_campaign=10.03.2017%20(1)>)

³² Users' tolerance for imperfections in products isn't blind however and depends on firsthand examination of the product and the reputation and persuasiveness of individual producer.

³³ Petroski 2010. p. ix

³⁴ Petroski 2010. p. 21

³⁵ Petroski 2010. p. 24

³⁶ Petroski 2010. p. 26

³⁷ Engineering Is Not Science - IEEE Spectrum. Downloaded from <https://spectrum.ieee.org/at-work/tech-careers/engineering-is-not-science>